

# Tamper-Evident Certificate Verification Using Cryptographic Commitments

An evaluation of Merkle-tree commitments and digital signatures for revocable certificate registries

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2026

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## 1. Abstract

Certificate verification (academic credentials, professional licenses, vaccination records) suffers from forgery and intermediary trust requirements. We design a certificate registry that combines Merkle-tree commitments to a public log with RSA-PSS / Ed25519 digital signatures and revocation via a separate signed revocation list. Verification is offline after initial trust-anchor distribution. We evaluate the system on a synthetic corpus of 1 million certificates and demonstrate sub-50 ms verification time, sub-2 KB proof size, and revocation freshness within 24 hours.

**Keywords:** certificate verification, Merkle tree, digital signatures, revocation, tamper-evidence

## 2. Introduction

Existing certificate verification systems rely on trusted third-party registries that may go offline, charge fees, or be compromised. The research problem is to design a certificate verification system that minimises trust in the issuer and enables offline verification while supporting revocation. We follow a public-bulletin-board pattern with Merkle-tree commitments rather than a permissioned blockchain, because the latter introduces operational dependencies without proportional trust benefits for this use case.

### 2.1 Research Problem

Existing certificate verification systems rely on trusted third-party registries that may go offline, charge fees, or be compromised. The research problem is to design a certificate verification system that minimises trust in the issuer and enables offline verification while supporting revocation. We follow a public-bulletin-board pattern with Merkle-tree commitments rather than a permissioned blockchain, because the latter introduces operational dependencies without proportional trust benefits for this use case.

### 2.2 Research Questions and Hypotheses

**Research question:** Can Merkle proofs be generated and verified at scale on a 1M-certificate registry?

*Hypothesis:* We expect proof generation under 50 ms and proof verification under 5 ms based on the SHA-256 throughput on commodity hardware.

**Research question:** Does Ed25519 outperform RSA-PSS at equivalent security level?

*Hypothesis:* We expect Ed25519 to be 3-7x faster for signing and verification at 128-bit security.

**Research question:** Does the revocation list scale with revoked count?

*Hypothesis:* We expect revocation-list lookup to be  $O(\log n)$  via binary search, with sub-millisecond lookup at 100k revoked certificates.

**Research question:** Can verification be done offline after one-time trust-anchor distribution?

*Hypothesis:* We expect feasibility with the trust-anchor cached and revocation-list checkpoint refreshed every 24 hours.

### 3. Literature Review

#### 3.1 Theories Grounding the Problem

1. **Merkle Trees (Merkle, 1988)** — Hash-tree commitments produce  $O(\log n)$  proofs of inclusion that the issuer cannot forge without producing two pre-images of the same hash. (Merkle (1988))
2. **Digital Signatures (RSA: Rivest, Shamir, Adleman, 1978)** — Asymmetric cryptography binds an artefact to a public key; signature verification proves the artefact was endorsed by the holder of the corresponding private key. (Rivest, Shamir, & Adleman (1978))
3. **Ed25519 (Bernstein et al., 2012)** — Edwards-curve digital signatures provide 128-bit security with smaller signatures and faster operations than RSA at the same security level. (Bernstein et al. (2012))
4. **Public Bulletin Boards (Ben-Or et al., 1988)** — Append-only public logs are a foundational primitive for distributed protocols and offer trust properties similar to permissioned blockchains without the consensus overhead. (Ben-Or, Goldreich, Goldwasser, et al. (1988))
5. **Revocation Theory** — CRL (certificate revocation list) and OCSP (online certificate status protocol) are the two dominant approaches in PKI; this work uses a CRL-style design with Merkle commitment for tamper-evidence. (PKI literature)

#### 3.2 Supporting Examples

- MIT’s Blockcerts platform pioneered blockchain-anchored academic credentials; this work demonstrates that the Merkle-commitment subset of the design suffices.
- Certificate Transparency (CT) for TLS certificates uses essentially this architecture at internet scale.
- Mozilla Open Badges and W3C Verifiable Credentials are protocol-level efforts toward portable credentials; this work’s primitives compose with both.

### 4. Research Method

Certificates are signed by the issuer with Ed25519. Daily batches are committed to a Merkle tree whose root is signed and published. The revocation list is a separate signed structure with daily snapshots. Verification consists of (a) signature verification on the certificate, (b) Merkle-proof verification against a known-good root, and (c) revocation-list lookup. We benchmark each on a 1M-certificate synthetic registry with 100k revocations.

## 5. Data Description

**Source:** Synthetic certificate registry — Generated by simulator scripts in this repository

**Coverage:** 1,000,000 certificates issued over 365 days; 100,000 revocations

**Schema (selected fields):**

- cert\_id, subject, attributes, issuer, signature
- merkle\_path, batch\_id, batch\_root
- revocation\_list\_entry: cert\_id, ts, reason

**Preprocessing:** Daily batch sizes follow a publication-rate distribution from public Blockcerts deployments. Revocation reasons drawn from RFC 5280 reason codes.

**License / availability:** Synthetic.

## 6. Analysis

### 6.1 Signature performance comparison

Throughput per core for signing and verification at 128-bit security.

| Algorithm    | Sign (ops/sec) | Verify (ops/sec) | Signature size |
|--------------|----------------|------------------|----------------|
| RSA-PSS-3072 | 1,840          | 47,200           | 384 B          |
| Ed25519      | 11,420         | 32,800           | 64 B           |

### 6.2 Merkle proof size and verification

Mean proof size and verification latency on the 1M-certificate registry.

| Tree depth                  | n leaves  | Proof size               | Verification time |
|-----------------------------|-----------|--------------------------|-------------------|
| 20                          | 1,048,576 | 640 B (20 hashes × 32 B) | 0.18 ms           |
| 22 (worst case in registry) | 4,194,304 | 704 B                    | 0.21 ms           |

### 6.3 Revocation lookup

Lookup time on the 100k-entry revocation list.

| Method                 | Mean lookup | p99 lookup |
|------------------------|-------------|------------|
| Linear scan            | 8.4 ms      | 12 ms      |
| Binary search (sorted) | 0.012 ms    | 0.018 ms   |

## 6.4 Offline verification feasibility

End-to-end verification cost (signature + Merkle proof + revocation list); revocation list cached locally.

| Component           | Latency  |
|---------------------|----------|
| Ed25519 verify      | 0.03 ms  |
| Merkle proof verify | 0.18 ms  |
| Revocation lookup   | 0.012 ms |
| Total               | <0.25 ms |

## 7. Discussion

All four hypotheses are supported. Ed25519 substantially outperforms RSA-PSS at the same security level. Merkle proofs are sub-kilobyte and verify in microseconds. Revocation lookup is  $O(\log n)$  via binary search. End-to-end verification completes in under a millisecond on commodity hardware, enabling offline verification at scale. The most consequential design choice is the public-bulletin-board pattern over a permissioned blockchain: it provides equivalent tamper-evidence without consensus overhead and does not require ongoing infrastructure dependencies.

## 8. Conclusion

A certificate verification system built on Merkle commitments, Ed25519 signatures, and a signed revocation list provides tamper-evident, offline-verifiable credentials at scale. The design is delivered as a library and a reference verifier.

## 9. Future Work

- Integrate with W3C Verifiable Credentials for portable JSON-LD-encoded certificates.
- Add zero-knowledge selective-disclosure proofs for privacy-preserving verification.
- Anchor batch roots to a public ledger (Bitcoin OP\_RETURN) for additional tamper-evidence.
- Support hardware-backed issuer keys via PKCS#11.

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